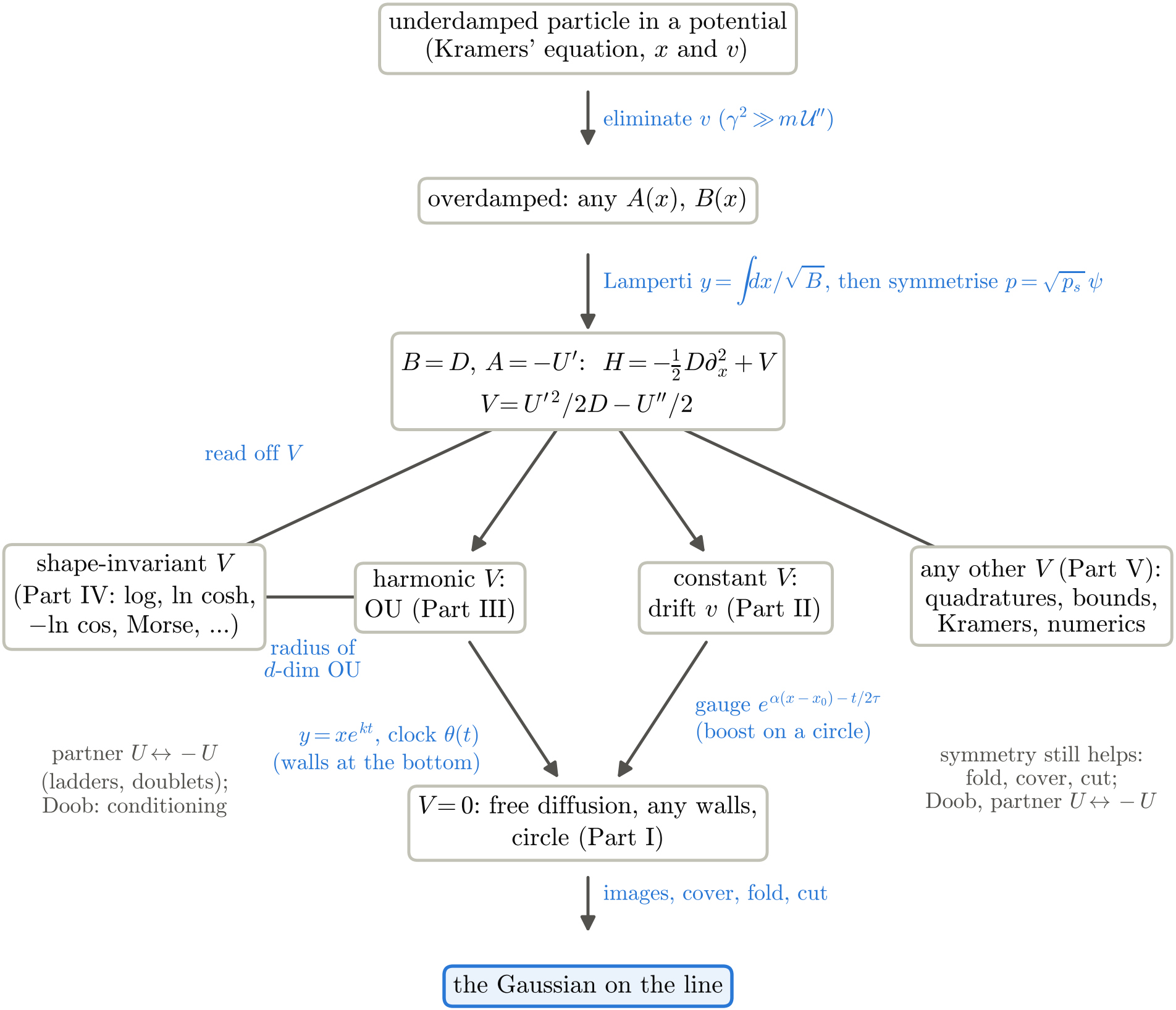
underdamped particle in a potential (Kramers' equation, x and v)
eliminate v ($\gamma^2 \gg m\mathcal{U}''$)
overdamped: any $A(x)$, $B(x)$
Lamperti $y = \int dx/\sqrt{B}$, then symmetrise $p = \sqrt{p_s}\,\psi$
$B = D$, $A = -U'$: $H = -\frac{1}{2}D\partial_x^2 + V$
$V = U'^2/2D - U''/2$
read off V
shape-invariant V (Part IV: log, ln cosh, $-$ln cos, Morse, ...)
harmonic V: OU (Part III)
constant V: drift v (Part II)
any other V (Part V): quadratures, bounds, Kramers, numerics
radius of d-dim OU
$y = xe^{kt}$, clock $\theta(t)$ (walls at the bottom)
gauge $e^{\alpha(x-x_0)-t/2\tau}$ (boost on a circle)
partner $U \leftrightarrow -U$ (ladders, doublets); Doob: conditioning
symmetry still helps: fold, cover, cut; Doob, partner $U \leftrightarrow -U$
$V = 0$: free diffusion, any walls, circle (Part I)
images, cover, fold, cut
the Gaussian on the line